



CLOSED CELL SPRAY FOAM

Dynamo ECO 2000

Medium-Density Closed Cell Spray Polyurethane Foam

Manufactured by **Dynamo Polyurethane Systems LLC** · 2113 Harwood Rd., Suite 309 #900, Bedford, Texas 76021 ·
Country of origin: **United States**

Dynamo ECO 2000 is a two-component, medium-density closed cell spray polyurethane foam. Closed cell foam is the densest, highest-performing insulation we install. It does four jobs at once: it insulates, it air-seals, it acts as a vapor retarder, and it adds rigidity to the framing it is sprayed against. We specify it where moisture control and maximum R-value per inch matter most.

R-7.5 per inch ASTM C518	>97.5% closed cell content ASTM D2856	2.0-2.1 lb/ft ³ core density ASTM D1622	1 hr / 2 hr re-entry / re-occupancy at 10 ACH
---------------------------------------	---	---	--

WHY WE SPECIFY IT

1 Highest R-value per inch we install

At R-7.5 per inch, ECO 2000 delivers code-required performance in roughly half the depth of open cell foam. That matters in rafter bays and rim joists where cavity depth is limited.

2 Built-in vapor retarder

At a minimum thickness of 1.5 inches, ECO 2000 qualifies as a Class II vapor retarder under ICC and ASHRAE. No separate poly sheet needed in the assembly.

3 Air-seals the envelope

Air permeance under 0.002 L/s·m² means the foam itself stops air movement. Air leakage accounts for a large share of heat loss in older homes, and no amount of fiberglass fixes it.

4 Zero ozone-depleting blowing agents

ECO 2000 uses the latest generation of blowing agents with zero ozone depletion potential, and carries 22.7% combined renewable and recycled content in the finished foam.

HEALTH & INDOOR AIR QUALITY

- **Low-VOC formulation.** ECO 2000 is classified as a low-VOC product. At applicable ventilation rates it allows worker re-entry after 1 hour and job site re-occupancy after 2 hours — among the shortest windows available in closed cell foam.
- **VOC emissions tested.** Passed 25-hour VOC emissions testing to CAN/ULC-S774.
- **No mold or fungal growth.** Tested to ASTM C1338 with no growth. Closed cell foam does not absorb water (0.5% by volume), so it does not become a food source or a reservoir for moisture.
- **Our standard practice with children in the home.** Regardless of the manufacturer's 2-hour figure, we recommend families with young children, elderly members, or anyone chemically sensitive plan to be out of the house during spraying and for a full 24 hours afterward. The extra margin costs you nothing.

TECHNICAL PROPERTIES

PROPERTY	VALUE	TEST METHOD
R-Value at 1 inch	7.5	ASTM C518
Closed-cell content	>97.5%	ASTM D2856
Core density	2.0 - 2.1 lb/ft³	ASTM D1622
Air permeance at 75 Pa	< 0.002 L/s·m²	ASTM E2178
Tensile strength	33 psi (pass)	ASTM D1623
Compressive strength	26.4 psi	ASTM D1621
Water vapor transmission	0.83 perms at 1 in.	ASTM E96
Water absorption	0.5% by volume	ASTM D2842
Dimensional stability	-4.45% at 158°F / 97% RH	ASTM D2126
Fungi resistance	Pass — no growth	ASTM C1338
VOC emissions	25 hours — passed	CAN/ULC-S774
Spontaneous ignition temp.	> 850°F	ASTM D1929
Flame spread	Class I — < 15	ASTM E84
Smoke development	Class I — < 300	ASTM E84
Potential heat	1,989 Btu/ft² per inch	NFPA 259

CODE COMPLIANCE & CERTIFICATIONS

Code Compliance Research Report	Intertek CCRR-0491
Construction types approved	Type I, II, III, IV and V
Thermal barrier	NFPA 286 — Pass; compliant with IBC / IRC
Exterior wall systems	NFPA 285 — Pass
Attics & crawl spaces	ICC-ES AC377 Appendix X — Pass. Approved for use without a prescriptive ignition barrier or intumescent coating.
Vapor retarder	Qualifies as ICC / ASHRAE Class II at 1.5 in. minimum thickness

ENVIRONMENTAL & CONTENT

Finished foam renewable + recycled content
22.7%
Polyol recycled content
37.4%
Polyol renewable content
8%
Ozone-depleting blowing agents
None

MANUFACTURER DOCUMENTATION

Technical Data Sheet (TDS)	dynamosp.com → Products Tech Info → Dynamo
Safety Data Sheet — Part B resin	Dynamo SDS 2000 ALL B
Safety Data Sheet — Part A / MDI ISO	Dynamo SDS ISO A
Code Evaluation Report	Intertek CCRR-0491
Full TDS and SDS for every product on your project are available on request, at no charge, same day.	

COATING & ASSEMBLY NOTE

When an intumescent thermal barrier coating is specified over ECO 2000, the manufacturer's approved products and rates are: ICP FireShell F10E at 14 wet mils, DC315 at 18–20 wet mils, Flame Control 60/60A at 14 wet mils, and No-Burn Plus ThB